Final Insights

Insight 1: Trader Profitability Peaks During Greed Sentiment

Analysis of Closed PnL across different market sentiment regimes revealed that traders achieved the highest average profitability during Greed periods. The average Closed PnL during Greed was 87.89, compared to 50.05 during Fear, 25.42 during Extreme Greed, and 22.23 during Neutral conditions.

This indicates that positive market sentiment creates more favorable trading environments, likely due to stronger trends and improved market participation. Interestingly, profitability declines significantly during Extreme Greed, suggesting that excessive optimism may lead to poorer decision-making and increased risk exposure.

Average Closed PnL by Sentiment: Greed: 87.89 **|** Fear: 50.05 **|** Extreme Greed: 25.42 **|** Neutral: 22.23

Insight 2: Traders Increase Position Size During Extreme Greed

Position sizing behavior varies considerably across sentiment regimes. The highest average trade size was observed during Extreme Greed conditions, reaching approximately 5,660 USD per trade. This is substantially higher than the average trade size during Greed (3,183 USD) and Neutral (3,059 USD) periods.

The findings suggest that traders become more aggressive and confident when market sentiment reaches extreme optimism levels. However, despite larger positions, profitability does not improve, indicating inefficient risk-taking behavior.

Average Position Size by Sentiment: Extreme Greed: 5,660 **|** Fear: 5,260 **|** Greed: 3,183 **|** Neutral: 3,059

Insight 3: Greed Periods Exhibit Stronger Profit-Taking Behavior

Trade direction analysis shows a noticeable shift in trader behavior during Greed periods. Approximately 57.5% of trades were SELL orders during Greed sentiment, compared to 42.5% BUY orders.

This suggests that traders actively realize profits when markets are optimistic rather than continuously increasing exposure. Such disciplined profit-taking behavior may be one of the key reasons behind the higher profitability observed during Greed periods.

Buy vs Sell Distribution: Greed: BUY 42.49%, SELL 57.50% Fear: BUY 49.36%, SELL 50.64% Extreme Greed: BUY 48.42%, SELL 51.58% Neutral: BUY 49.08%, SELL 50.92%

Strategic Recommendations

1. Increase Participation During Greed Periods.
Historical performance indicates that Greed sentiment is associated with the highest average trader profitability.

2. Reduce Position Size During Extreme Greed.
Reducing exposure by 20–30% during Extreme Greed may help protect capital and reduce losses caused by overconfidence.

3. Implement Sentiment-Aware Risk Management.
Position sizing, trade frequency, and exposure limits can be dynamically adjusted based on prevailing sentiment conditions.